

MISM 6210 - FINAL REPORT

Executive Summary

In the evolving landscape of air travel, understanding the factors that influence airfare trends is essential for travelers, airline planners, and policymakers alike. Our project investigates how domestic airfares across the United States have changed between 2018 and 2024, with a particular focus on the COVID-19 pandemic's disruption, inflationary effects, and regional disparities. Using Tableau as our primary visualization tool, we analyzed airport-level fare data—both nominal and inflation-adjusted—to reveal patterns in pricing, passenger volume, and variability across states and cities.

Research Problem: Why It Matters

Airfare is a highly dynamic cost influenced by travel demand, global events, and economic shifts. The COVID-19 pandemic caused a massive disruption in air travel, followed by price rebounds that vary by region and airport. Our research question is:

“How have average domestic airfares evolved across U.S. regions over time—especially through the pandemic—and what factors (traffic volume, inflation, location) influence these shifts?”

This problem has implications for economic equity, tourism recovery, and transportation affordability. By analyzing nominal and real fares across states, we aim to expose hidden cost burdens and pricing trends that might otherwise go unnoticed.

Data & Methodology

The dataset spans the years 2018 to 2024 and includes detailed information at the airport level, such as average fare, inflation-adjusted fare, city, state, airport code, and annual data. A notable feature is the “Passengers (10% sample)” column, which provides a representative sample of total passenger traffic per airport. Although this does not reflect full traffic counts, it serves as a reliable proxy for comparing relative airport activity. The volume sample allows for meaningful comparisons of fare trends in relation to airport busyness, which is particularly valuable in identifying whether high-traffic hubs offer cost advantages to travelers. Our visualizations were created in Tableau and include descriptive, comparative, and diagnostic dashboards that focus on fare trends, regional changes, inflation-adjusted cost differences, and variability in ticket pricing over time.

Two core dashboards were designed to convey temporal and spatial dynamics. The first, a 2024 state-level dashboard, excluded Alaska from the map visualization. This design decision was made because Alaska's significantly higher fares skewed the color scale, which made it difficult to observe differences among other states. By removing Alaska, we improved clarity and allowed regional patterns to emerge more distinctly. The second dashboard used a dumbbell chart comparing 2018, 2020, and 2024, highlighting fare evolution before, during, and after the

COVID-19 pandemic. This three-point temporal comparison effectively captures the trajectory of fare disruption and recovery.

Key Visualizations and Insights

1. Passenger Volume vs Average Fare by Year - The scatter plot shows a clear inverse relationship between passenger volume and average fare. Airports with higher traffic tend to have significantly lower average fares, reflecting the influence of scale economies and increased competition. Conversely, airports with very low passenger counts often have much higher fares, indicating pricing inefficiencies or limited availability of carriers. This trend underscores how traffic volume is a major factor in airfare affordability.

2. Year-on-Year Variability of Average Airfares (Top 10 U.S. States) - The box plot illustrates how fare consistency varies across states over time. From 2018 to 2024, some states experienced increasing variability, with wider interquartile ranges and more outliers in recent years. This suggests that airfare pricing has become less predictable for travelers, possibly due to fluctuating demand, supply chain disruptions, or shifting airline strategies in response to the pandemic.

3. Top States' Fare Patterns: A 7-Year Overview - The line graph compares fare trends for ten states across seven years. States like Idaho and Kentucky showed a sharp rise in fares, while others like Hawaii and Vermont maintained more gradual increases. The visual reveals long-term fare trajectories and helps identify regions where pricing has become significantly more expensive, making it easier to forecast future travel costs based on historical behavior.

4. Hidden Inflation Impact on Airfares Across U.S. States - The choropleth map visualizes the difference between nominal and inflation-adjusted fares across states. Several states, particularly in the West and Northeast, show hidden inflation gaps exceeding \$100, meaning that the real cost of travel is significantly higher than what passengers perceive. This highlights the importance of looking beyond nominal fares to understand true affordability.

5. Average Airfare by State – 2024 - This filled map displays average fares across states in 2024, with Alaska standing out as the most expensive state at \$711. Western states like Idaho, Utah, and California also show high averages, while Southern states generally remain more affordable. The visualization emphasizes stark regional differences in travel costs that can influence both consumer behavior and airline route planning.

6. Top 20 Most Expensive U.S. Cities by Inflation-Adjusted Airfare (2024) - The horizontal bar chart ranks cities by their inflation-adjusted fares in 2024, with St. Paul, Unalakleet, and Sandpoint topping the list. Many of these cities are remote or less accessible, which likely contributes to their elevated fare levels. The chart reveals how geographical isolation and limited competition drive up the real cost of air travel in specific locations.

7. Are High-Traffic Airports More Affordable - This scatter plot ranks airports by fare and shows that higher-ranked (i.e., lower traffic) airports tend to have higher fares. A subtle regression line indicates that as airport traffic increases, fares tend to decrease. This supports the conclusion that larger, busier airports offer more competitive pricing due to scale, route options, and airline presence.

8. How Airfares Shifted Across 2024 – Quarterly Comparison - The stacked bar chart compares quarterly fares across all U.S. states in 2024. Seasonal peaks and valleys can be observed, with Q2 and Q4 generally showing higher prices. This chart provides insight into intra-annual pricing patterns, helping stakeholders identify periods of heightened demand or operational strain that lead to fare surges.

9. Pre vs. Post-Pandemic Airfare Trends – Dumbbell Chart - The dumbbell chart compares fares from 2018, 2020, and 2024 for selected airports, effectively illustrating fare changes over three distinct phases: pre-pandemic, during the pandemic, and post-pandemic. Remote airports like Guam and Pago Pago showed large fluctuations, while others demonstrated more stable recovery. This visualization captures the uneven pace of rebound in air travel markets.

10. How U.S. Airfares Shifted (2018–2019 vs. 2024 Average) - This choropleth map highlights how average fares have changed from the 2018–2019 baseline to 2024. Western states like Utah, Nevada, and Arizona show the highest increases, exceeding 25%, while some states in the Midwest saw minimal change or even slight declines. The map underscores geographic inequality in post-COVID fare recovery.

11. From Dip to Rise: What Changed in Airfare Trends? - The line area chart reveals the broader fare trend from 2018 to 2024. It shows a steep drop to \$359 in 2020 due to COVID-19, followed by a strong rebound through 2021–2024, reaching \$478. This pattern illustrates how global disruptions can cause sudden fare shifts and how recovery tends to overshoot pre-crisis levels in response to pent-up demand and inflation.

Conclusions

The analysis clearly indicates that fare trends cannot be fully understood through nominal pricing alone. Inflation-adjusted fare analysis reveals that the real cost of travel has increased more significantly than most travelers realize. Additionally, regional recovery patterns show that certain states have rebounded much faster or more aggressively than others, resulting in affordability disparities across the country. Airports with higher passenger volume tend to offer more stable and lower fares, while smaller or remote airports exhibit greater price volatility. Moreover, fare variability has increased year over year in some states, making it harder for travelers to anticipate or plan for stable pricing.

Recommendations

Based on our findings, we recommend that travelers and transportation planners begin tracking inflation-adjusted fare trends alongside nominal pricing to better understand true cost changes. Tourism boards and policy agencies should consider regional inflation gaps when making decisions about travel promotion or accessibility initiatives. Airlines can also benefit from this analysis by optimizing pricing strategies based on traffic volume and inflation-adjusted demand. Lastly, transparency tools and public dashboards should present both nominal and real fare data to give consumers a clearer picture of what they're paying for.

Limitations

- Passenger Volume Sampling: The dataset includes only a 10% sample of total passenger traffic, which limits the precision of volume-based insights.
- Lack of Route and Carrier Detail: There is no information on specific routes (origin–destination pairs), airline carriers, or travel class, which restricts analysis of airline-specific pricing behavior.
- Missing External Influences: Key pricing influencers such as weather events, fuel prices, promotional campaigns, and customer satisfaction data are not included, which could
- affect short-term fare trends.

Future Work

- Enhanced Data Integration: Incorporate real-time traffic, booking behavior, and travel class data to improve the accuracy of fare trend analysis.
- Factor-Based Modeling: Include variables such as fuel pricing, route competition, and airline capacity to build predictive models with greater real-world relevance.
- Behavioral Segmentation and Disruption Analysis: Segment travelers by booking frequency or demographics, and integrate weather and disruption data to develop robust, traveler-centric pricing insights.